Lesson 13.1 - The Least Squares Method

> **Video:** Linear Regression Using Least Squares Method - Line of Best Fit Equation by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

Goal

Find the straight line that **best fits** a set of points that don't lie exactly on a line. The **least squares method** chooses the line that makes the points as close to it as possible: it minimizes the sum of the squared vertical distances.

The line

Algebra	Statistics	Meaning
$y = mx + b$	$\hat{y} = b_0 + b_1 x$	
m	b_1	slope
b	b_0	y-intercept

Formulas

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}, \qquad b = \frac{\sum y - m\sum x}{n}$$

n = number of data points. Keep m unrounded when you calculate b.

Example

x	y	xy	x^2
1	1.5	1.5	1
2	3.8	7.6	4
3	6.7	20.1	9
4	9.0	36.0	16
5	11.2	56.0	25
6	13.6	81.6	36

x	y	xy	x^2
7	16.0	112.0	49
Sum	**61.8**	**314.8**	**140**

$\sum x = 28,\ \sum y = 61.8,\ \sum xy = 314.8,\ \sum x^2 = 140,\ n = 7$

Slope:

$$m = \frac{7(314.8) - 28(61.8)}{7(140) - 28^2} = \frac{2203.6 - 1730.4}{980 - 784} = \frac{473.2}{196} \approx 2.414286$$

Intercept:

$$b = \frac{61.8 - 2.414286(28)}{7} = \frac{61.8 - 67.6}{7} = \frac{-5.8}{7} \approx -0.828571$$

Line of best fit:

$$\boxed{y \approx 2.41x - 0.83}$$

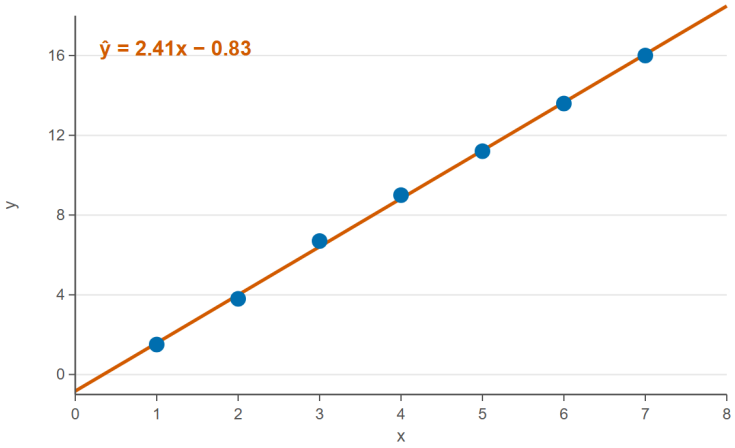
Checking the fit

x	**Predicted** $y = 2.41x - 0.83$	**Actual** y
2	3.99	3.8
5	11.22	11.2
7	16.04	16.0

The predictions are very close to the data, so the line fits well.

Least squares: the line with the smallest sum of squared residuals

Residual = actual y − predicted y (the vertical gaps). The best-fit line makes the sum of their squares as small as possible.



x	y	predicted	residual
1	1.5	1.59	-0.09
2	3.8	4.00	-0.20
3	6.7	6.41	+0.29
4	9	8.83	+0.17
5	11.2	11.24	-0.04
6	13.6	13.66	-0.06
7	16	16.07	-0.07

sum of squared residuals ≈ 0.169

Any other line gives a larger sum.

The residuals are tiny here, so the red gaps on the plot are barely visible.

The least squares line passes very close to every point. The residuals (actual minus predicted) are all small, and their squares add up to about 0.17, which is the smallest possible for any straight line.

Doing it in Excel

With the x values in `B2:B8` and the y values in `C2:C8`:

Want	Formula	Result
Slope m	`=SLOPE(C2:C8, B2:B8)` (known **y**'s first, then known **x**'s)	2.414
Intercept b	`=INTERCEPT(C2:C8, B2:B8)`	−0.8286

Key takeaways

- Build a table with columns x, y, xy, x^2 and add up each column.
- $m = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$ and $b = \dfrac{\sum y - m\sum x}{n}$.
- Don't round m before calculating b.
- Excel's `SLOPE` and `INTERCEPT` give the same results in seconds.